

SOLANA SEEKER · SUBSCRIPTIONS STANDARD · CLOCK IN & CRYPTO WORLD'S FAIR

nuntius

**Grant it once in Seed Vault.
The chain holds the line.**

A capped, revocable, recurring payment you approve once, with a push receipt for every pull, including the ones the chain refused.

PROBLEM

Solana has a recurring-payments standard. Users have no place to see or stop it.

Live since 2 June 2026

The Foundation's Subscriptions program. Audited by Cantina. The same address on devnet and mainnet.

u64::MAX at the token level

Every grant makes a program PDA the token account's delegate for the maximum amount. The per-period cap lives inside the program.

Every integration is merchant-side

Plans, billing, collection. We found no wallet or app that shows the user these permissions, alerts when money moves, or ends them in one tap.

Sources: the program repository and CHANGELOG (read at commit 299dfea); twins survey in RESEARCH.md. "Merchant-side only" is a search result, not a census.

PRODUCT

Grant. Watch. Control.

Grant, in one sentence

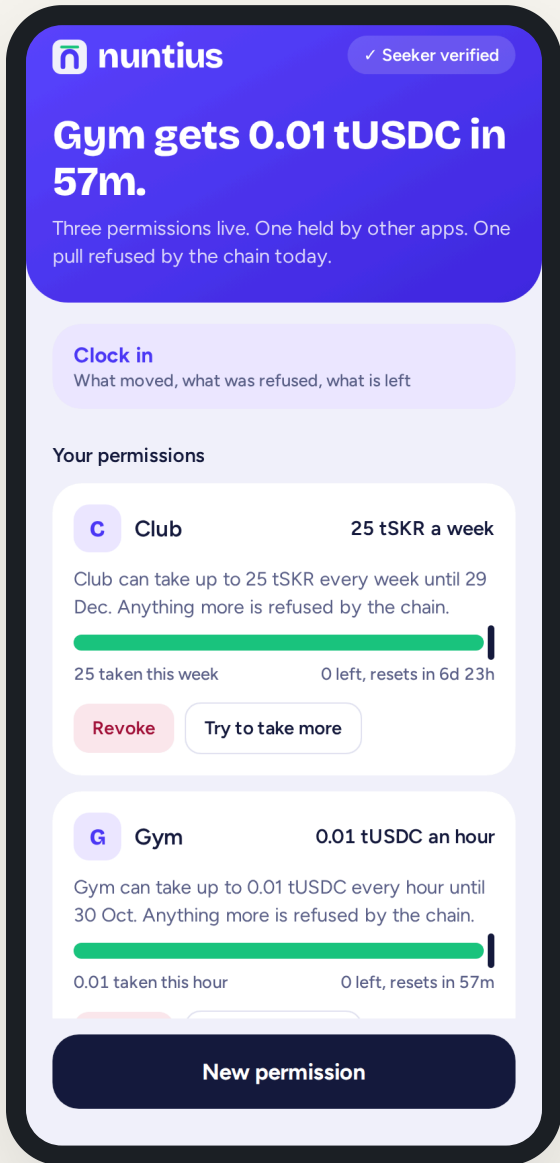
"Ana can receive up to 25 USDC every week, until 26 Dec." One Seed Vault approval. The first payment goes out at once, then one per period, with the phone in your pocket.

Watch, with a receipt for every pull

Every pull opens the on-chain proof. When anything asks above the cap, the program refuses it (0x190) and the phone says so.

Control every permission

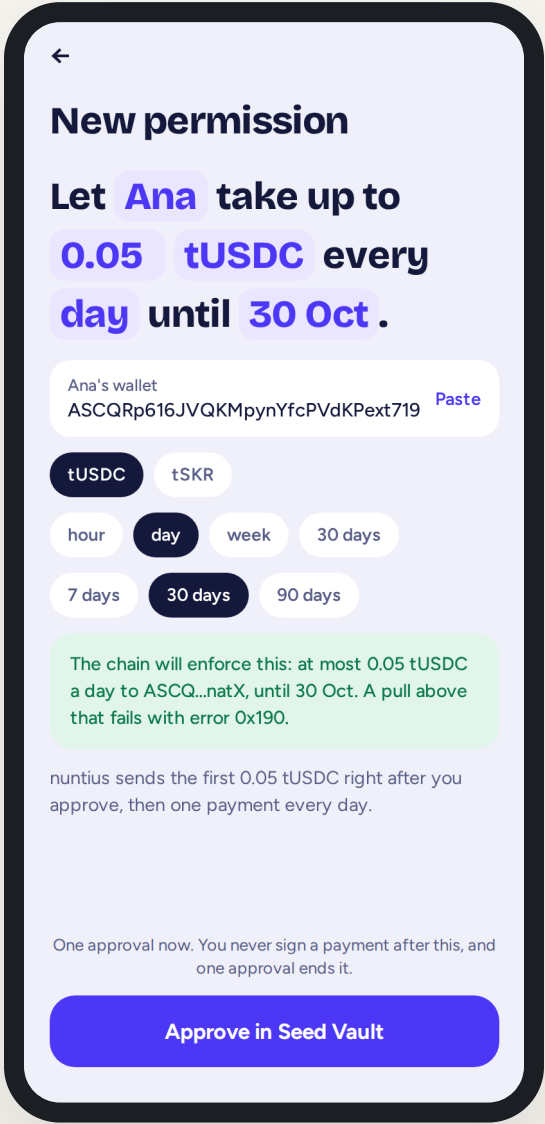
All Subscriptions delegations on the wallet, to nuntius or to any other app. Each ends with one approval.



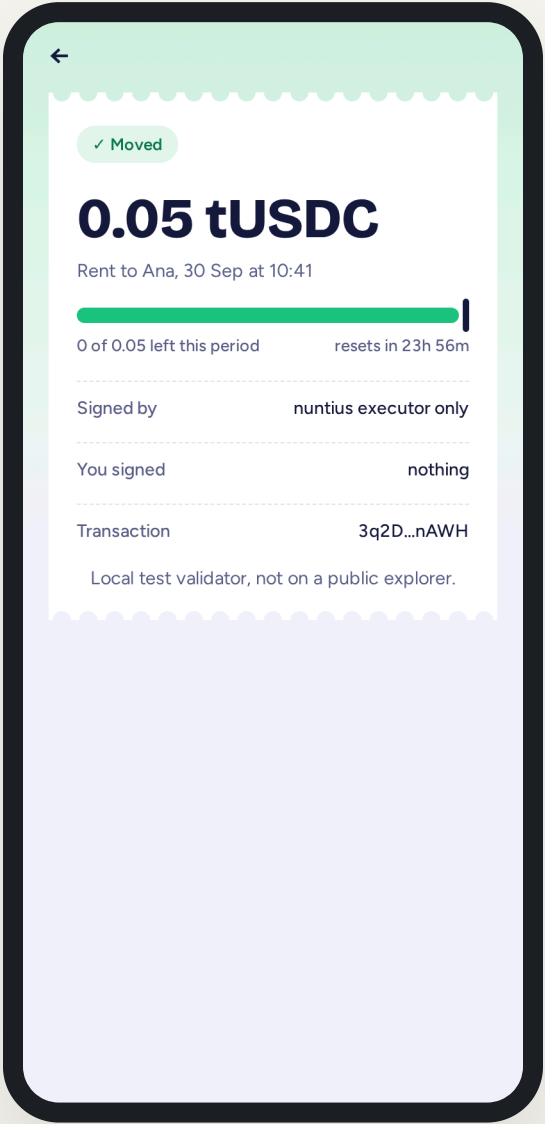
Permissions screen. Web build of the app against a local validator running the real program; test mint tUSDC.

THE PHONE

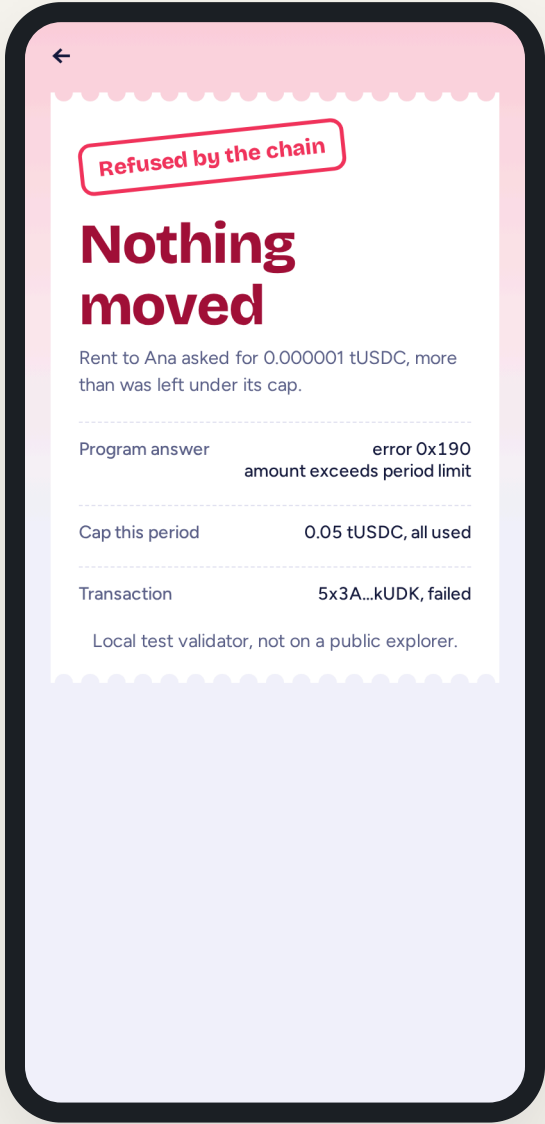
What the user sees



The sentence is the server's parse of the exact terms the transaction carries.



Pull receipt: what moved, what is left, the reset, the transaction.



Refused by the chain: 0x190. One base unit over is enough to be refused.

All three are the real app screens, rendered from its web build against a local validator (test mint tUSDC). On-device recording: see the demo video.

Authority and delegation in one transaction

initSubscriptionAuthority

Creates the (user, mint) authority PDA and makes it the token account's delegate. Idempotent when it already exists.

+

createRecurringDelegation

Amount per period, period length, expiry. It pins the authority's `init_id`, or `UNKNOWN_INIT_ID` when the authority is created in the same slot (program 0.5.0, #206).

=

1 signer: the user

One Seed Vault sheet. Revoke is the same: `revokeDelegation` + `revokeSubscriptionAuthority` in one transaction when it is the last delegation on that mint.

Measured

The localnet test asserts the grant's required signers are [owner], the delegation carries the exact terms, and the token account delegate is the authority at 18446744073709551615.

Not yet measured

Seed Vault signing the one-transaction grant on the Seeker, on mainnet. The two-transaction grant was signed there on 22 Sep. The one-transaction version is still to be recorded (see "Next").

It already ran on mainnet from the phone

STEP	RESULT	SIGNATURE
initSubscriptionAuthority · createRecurringDelegation	signed by Seed Vault, Seeker SM02E4060327059	bBNbUgwh...qGS2Sks · 5r7YnGrb...FUGinugMw
transferRecurring ×4	17,000 base units moved; one signer, the delegatee	59Zn57sY... 5kSbpeH6... 3utaq3gm... 41emXYEH...
Pull of 2× the cap	custom program error 0x190, landed	3RpcCkkL...PmaZpWyR7
One unit after the cap was used	0x190, landed	5ryxPMPj...oQ1TKAjsDpd
Period rolled (60 s)	pull allowed again, no new user signature	see README
revokeDelegation · revokeSubscriptionAuthority	both PDAs closed; delegate: none	Vh2yr9VT... · 2doDFNXU...
Pull after revocation	fails: account gone	cfESoUjx...uSqzEzTtM5GPzm

Cap 10,000 base units (0.01 USDC) per 60 s. The delegator ended 112,000 lamports down; both rents came back on revoke. Full signatures and Explorer links: README.md and JUDGE_GUIDE.md.

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tests pass against **solana-foundation/subscriptions** @ **364a419**, the commit its CHANGELOG names as the mainnet release, in a local validator (Agave 3.1.10).

sha256 31309d42...a30974648 · scripts/localnet.sh
reproduces it

- One-signature grant; second mandate reuses the authority with the real `init_id`
- Delegatee-only pulls; over-cap and one-unit-over refused with `{"Custom":400}`
- Executor: pulls, period roll and a second pull, a real 0x190, notices a revoke
- Guard: receipts for a foreign delegatee's pulls and refusals; flags new grants
- HTTP API end to end, driven exactly as the app drives it
- Revoke of the last delegation leaves `delegate: none`

Also: 24 server unit tests, 6 app-logic tests, typecheck, lint and prettier clean. Whether the local binary is byte-identical to mainnet has not been measured yet.

An executor that cannot double-pull

Idempotent per period

One ledger row per (delegation, period start), holding the signature *before* sending. A replacement is built only once the old blockhash can no longer land.

0x190 is final

Recorded as a refusal, pushed once, never retried. If the period's cap is already used, nothing is sent: no fees spent on a known no.

Flaky RPC, gone users

Exponential backoff with jitter. A closed delegation means revoked; past expiry means over. Each ends with one receipt.

12 unit tests on a simulated program (lost transactions, restarts, a send that throws after landing, max attempts). Logs are JSON with keys and tokens redacted.

STICKINESS

It comes to you, every day

A receipt on every pull

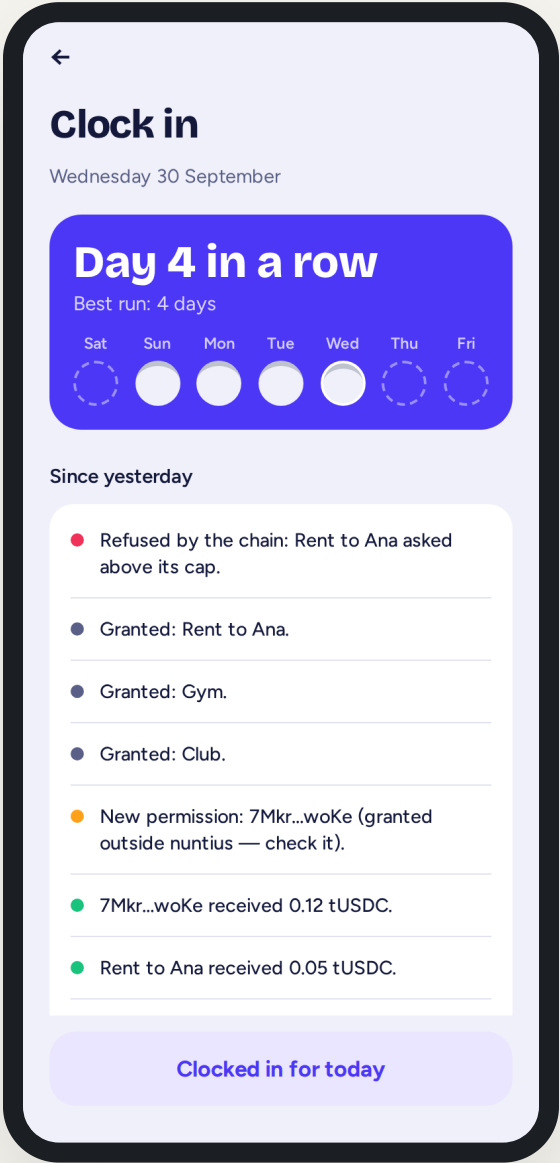
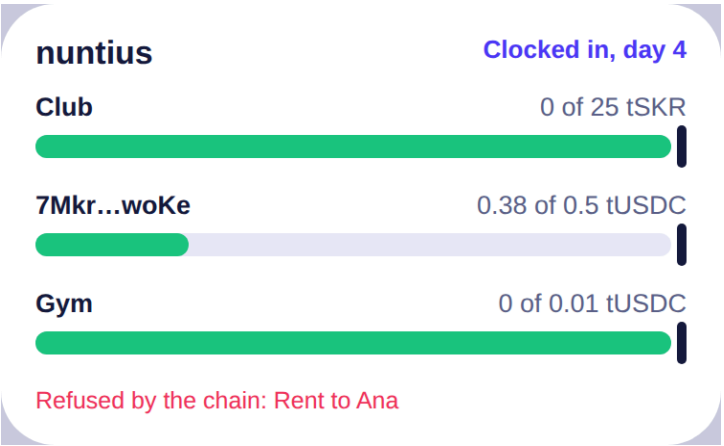
Money moves while the app is closed. The push is the reason to look.

Morning clock-in

At the hour you choose: what moved overnight, what was refused, what is left. Opening it is the clock-in, and it keeps a streak.

Home-screen widget

Cap left and time to reset for each permission.



Clock-in screen, web build against a local validator (test mint).

What the cap bounds, and what it does not

Bounded by the program

Amount per period (0x190); the delegatee is the only signer; whole-period resets; hard expiry; revoke closes it. Our server cannot raise any of it.

Not bounded by the cap

The token approval is **u64::MAX**. The program is **upgradeable** (Squads multisig, per its ADR-004). The **destination** is bound by nuntius, not by the chain.

Audit scope

The one-transaction path (#206) is inside Cantina's audited baseline d6b3a5dc. The release commit is 8 commits past the audit-fix commit.

This codebase

No secrets in the repo; SIWS nonces are single-use; sessions expire in 30 days; the server has 0 npm advisories. Full table in SECURITY.md.

FIELD	STANDARD AUDITED PROGRAM	PER-PERIOD CAP	ONE SEED VAULT APPROVAL	RECEIPT PER PULL	OTHER APPS' PERMISSIONS	MAINNET
nuntius	yes	yes	yes (localnet-proven)	yes, incl. refused	yes	yes (22 Sep)
Veto (Clock In)	own Anchor program	total and daily limits	Seed Vault owner key	on-chain refusal records	no	devnet
Kin (Clock In)	plain SPL approve	per round	MWA	reminders	no	devnet
Loonie	yes	yes	web wallet	no	no	devnet
Explorer card	yes	read-only	—	no	read-only	yes

From each project's public README, read on 27 Sep 2026 (RESEARCH.md). nuntius "Mainnet" is the 22 Sep life cycle (two approvals). The one-approval grant and receipts are proven on localnet and are still to be recorded on mainnet.

Where it stands, honestly

Done

Mainnet life cycle from the phone (22 Sep). One-approval grant and revoke, executor, guard, tier gate, digest, streak and widget, all built and tested against the real program. Docs, threat model, judge guide.

Before submission

Record the one-approval grant on mainnet from the Seeker. Pass the device checklist for the widget, receipts and digest. Publish the APK release.

After

dApp Store listing. Talk to the merchants on the standard. A disclosed fee on mandated flow, with the guard free for everyone. No users or revenue yet.

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